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### **ELECTRICAL WIRE CONNECTION IN INTRALUMINAL ULTRASOUND IMAGING DEVICES, SYSTEMS, AND METHODS**

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#### **Abstract**

An intraluminal imaging device includes a flexible elongate member configured to be positioned within a body lumen of a patient. The flexible elongate member includes a plurality of coaxial cables. Each of the plurality of coaxial cables includes a conductive shield layer. The intraluminal imaging device also includes an ultrasound imaging assembly positioned at a distal portion of the flexible elongate member and in communication with the plurality of coaxial cables. The ultrasound imaging assembly includes a transducer array configured to obtain ultrasound data and a conductive pad. The conductive shield layer of each of the plurality of coaxial cables is mechanically and electrically coupled to the conductive pad. Associated devices, systems, and methods are also provided.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 17/793,082, filed Jul. 15, 2022, now U.S. Pat. No. 12,303,333, which is the national stage entry of International Application No. PCT/EP2021/050031, filed Jan. 5, 2021, which claims priority to and the benefit of U.S. Provisional Application No. 62/961,883, filed Jan. 16, 2020, each of which is incorporated by reference herein in its entirety.

### TECHNICAL FIELD

[0002] The present disclosure relates generally to ultrasound imaging systems, and in particular, to electrical wire connections for imaging catheters, such as intra-cardiac echocardiography (ICE) catheters.

### BACKGROUND

[0003] Diagnostic and therapeutic ultrasound catheters have been designed for use inside many areas of the human body. In the cardiovascular system, a common diagnostic ultrasound method is intraluminal ultrasound imaging with intra-cardiac echocardiography (ICE) being a specific example of intraluminal imaging. Typically, a single rotating transducer or an array of transducer elements is used to transmit ultrasound at the tips of the catheters. The same transducers are used to receive echoes from the tissue. A signal generated from the echoes is transferred to a console which allows for the processing, storing, display, or manipulation of the ultrasound-related data.

[0004] Intraluminal imaging catheters, such as ICE catheters (e.g., Siemens Acunav, St. Jude ViewFlex), are generally used to image heart and surrounding structures, for example, to guide and facilitate medical procedures, such as transseptal lumen punctures, left atrial appendage closures, atrial fibrillation ablation, and valve repairs. Commercially-available ICE catheters have distal ends which can be articulated by a steering mechanism located in a handle at the proximal end of the catheter. For example, an intraluminal imaging catheter such as an ICE catheter may be inserted through the femoral or jugular vein when accessing the anatomy and steered in the heart to acquire images necessary to the safety of the medical procedures.

[0005] ICE catheters must meet national and/or international requirements that stipulate a distal tip of a catheter must not separate when tensile pull loads of various magnitudes are applied depending primarily on the diameter of the catheter. This ensures that the catheter can be safely removed from a patient's body without a breakage causing the tip to be left within the heart or vasculature. Because ICE catheters are necessarily extremely narrow and the distal catheter assembly houses many transducers, circuits, and other electrical components, achieving the necessary tensile strength in an extremely confined space is difficult.

### SUMMARY

[0006] Embodiments of the present disclosure are directed to electrical wire connections for imaging catheters, such as intra-cardiac echocardiography (ICE) catheters. There is a need for a simple, compact, stronger method and apparatus to connect the ultrasound imaging assembly of an ICE catheter to the elongate wires that carry it. To achieve a stronger mechanical coupling, a mass of solder can be applied to form a mechanical and electrical bond between the shield layers of a number of coaxial cables and a grounding pad positioned on an interposer within an ultrasound imaging assembly. The center conductors of these coaxial cables may be bonded to corresponding conductive pads elsewhere on the interposer and may provide signal communication between the

ultrasound imaging assembly and a computer. Single conductor cables may be connected to the interposer at other places within the ultrasound imaging assembly. The bonding of shield layers of coaxial cables to a grounding pad on the interposer provides a common ground signal relative to all center conductor signals of the coaxial cables and provides a strong physical connection between the catheter cables and the ultrasound imaging assembly making it easier to achieve the necessary tensile strength of intra-cardiac echocardiography (ICE) catheters.

[0007] According to an exemplary aspect of the present disclosure, an intraluminal imaging device is provided. The intraluminal imaging device comprises a flexible elongate member configured to be positioned within a body lumen of a patient, the flexible elongate member comprising a plurality of coaxial cables, wherein each of the plurality of coaxial cables comprises a conductive shield layer; and an ultrasound imaging assembly positioned at a distal portion of the flexible elongate member and in communication with the plurality of coaxial cables, the ultrasound imaging assembly comprising: a transducer array configured to obtain ultrasound data; and a first conductive pad, wherein the conductive shield layer of each of the plurality of coaxial cables is mechanically and electrically coupled to the first conductive pad.

[0008] In some aspects, the intraluminal imaging device further comprises: a solder mass positioned over the first conductive pad, wherein the conductive shield layer of each of the plurality of coaxial cables is mechanically and electrically coupled to the first conductive pad and to one another via the solder mass. In some aspects, the solder mass is positioned around a perimeter of the plurality of coaxial cables. In some aspects, the solder mass is configured to support a tensile load of at least 15 N. In some aspects, each of the plurality of coaxial cables comprises an insulating jacket around the conductive shield layer, the conductive shield layer comprises an exposed portion without the insulating jacket, and corresponding dimensions of the exposed portion of the conductive shield layer and the first conductive pad are equal. In some aspects, each of the plurality of coaxial cables comprises a center conductor and an insulation layer around the center conductor, the conductive shield layer is positioned around the insulation layer, and the solder mass is positioned between the conductive shield layer and the insulation layer. In some aspects, the plurality of coaxial cables comprises a first row of coaxial cables and a second row of coaxial cables, the first row of coaxial cables is positioned over the first conductive pad, and the second row is positioned over the first row of coaxial cables. In some aspects, the solder mass is positioned between the first row of coaxial cables and the second row of coaxial cables. In some aspects, each of the plurality of coaxial cables is spaced apart, wherein the solder mass is positioned between each of the plurality of coaxial cables. In some aspects, the first conductive pad comprises an electrical ground for the plurality of coaxial cables. In some aspects, each of the plurality of coaxial cables comprises a center conductor, wherein the ultrasound imaging assembly comprises a plurality of second conductive pads, the center conductor of each of the plurality of coaxial cables is mechanically and electrically coupled to a corresponding one of the plurality of second conductive pads, and the center conductor is configured to carry electrical signals to and from the ultrasound imaging assembly. In some aspects, the flexible elongate member comprises a plurality of single conductor cables positioned over the plurality of coaxial cables, the ultrasound imaging assembly comprises a plurality of third conductive pads configured to be mechanically and electrically coupled to the plurality of single conductor cables, and the plurality of single conductor cables are configured to carry the electrical signals to and from the ultrasound imaging assembly. In some aspects, the flexible elongate member comprises a catheter configured to be positioned within a heart of the patient. In some aspects, the ultrasound imaging assembly further comprises a circuit board in communication with the transducer array, the first conductive pad positioned on a surface of the circuit board.

[0009] According to an exemplary aspect of the present disclosure, a system is provided. The system comprises an intraluminal imaging device, comprising: a flexible elongate member configured to be positioned within a body lumen of a patient, the flexible elongate member

comprising a plurality of coaxial cables, wherein each of the plurality of coaxial cables comprises a conductive shield layer; and an ultrasound imaging assembly positioned at a distal portion of the flexible elongate member and in communication with the plurality of coaxial cables, the ultrasound imaging assembly comprising: a transducer array configured to obtain ultrasound data; and a conductive pad, wherein the conductive shield layer of each of the plurality of coaxial cables is mechanically and electrically coupled to the conductive pad; and a computer in communication with the intraluminal imaging device and configured to generate an ultrasound image based on the ultrasound data.

[0010] Additional aspects, features, and advantages of the present disclosure will become apparent from the following detailed description.

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## **Description**

### **BRIEF DESCRIPTION OF THE DRAWINGS**

[0011] Illustrative embodiments of the present disclosure will be described with reference to the accompanying drawings, of which:

[0012] FIG. 1A is a schematic diagram of an intraluminal imaging system according to embodiments of the present disclosure.

[0013] FIG. 1B is a schematic diagram of a catheter according to embodiments of the present disclosure.

[0014] FIG. 2 is a perspective view of a catheter cable according to embodiments of the present disclosure.

[0015] FIG. 3A is a cross-sectional view of a catheter body according to embodiments of the present disclosure.

[0016] FIG. 3B is a cross-sectional view of a catheter cable according to embodiments of the present disclosure.

[0017] FIG. 4 is a perspective view of a distal ultrasound assembly of an intraluminal imaging device according to embodiments of the present disclosure.

[0018] FIG. 5 is a flow chart diagram of a method of assembling an intraluminal imaging device according to an embodiment of the present disclosure.

[0019] FIG. 6A is a top view of an interposer before catheter cables are connected according to an embodiment of the present disclosure.

[0020] FIG. 6B is a cross-sectional view of the interposer along section line 6B-6B in FIG. 6A before catheter cables are connected according to an embodiment of the present disclosure.

[0021] FIG. 7 is a perspective view of a coaxial cable with each layer partially exposed according to an embodiment of the present disclosure.

[0022] FIG. 8A is a top view of the interposer of FIG. 6A with a first row of coaxial cables connected according to an embodiment of the present disclosure.

[0023] FIG. 8B is a cross sectional view of the interposer along section line 8B-8B in FIG. 8A with a first row of coaxial cables connected according to an embodiment of the present disclosure.

[0024] FIG. 9A is a top view of the interposer of FIG. 6A with first and second rows of coaxial cables connected according to an embodiment of the present disclosure.

[0025] FIG. 9B is a cross-sectional view of the interposer along section line 9B-9B in FIG. 9A with first and second rows of coaxial cables connected according to an embodiment of the present disclosure.

[0026] FIG. 10A is a top view of the interposer of FIG. 6A with first and second rows of coaxial cables connected, and a third top row of center conductor cables positioned above coaxial cables according to an embodiment of the present disclosure.

[0027] FIG. 10B is a cross-sectional view of an interposer with first and second rows of coaxial

cables connected, and a third top row of center conductor cables positioned above coaxial cables according to an embodiment of the present disclosure.

[0028] FIG. **11** is a perspective view of an interposer with first and second rows of coaxial cables connected, and a third top row of center conductor cables positioned above coaxial cables according to an embodiment of the present disclosure.

[0029] FIG. **12** is a schematic diagram of a processor circuit, according to an embodiment of the present disclosure.

#### DETAILED DESCRIPTION

[0030] For the purposes of promoting an understanding of the principles of the present disclosure, reference will now be made to the embodiments illustrated in the drawings, and specific language will be used to describe the same. It is nevertheless understood that no limitation to the scope of the disclosure is intended. Any alterations and further modifications to the described devices, systems, and methods, and any further application of the principles of the present disclosure are fully contemplated and included within the present disclosure as would normally occur to one skilled in the art to which the disclosure relates. For example, while the present disclosure is described in terms of intraluminal imaging, it is understood that it is not intended to be limited to this application. In particular, it is fully contemplated that the features, components, and/or steps described with respect to one embodiment may be combined with the features, components, and/or steps described with respect to other embodiments of the present disclosure. For the sake of brevity, however, the numerous iterations of these combinations will not be described separately.

[0031] FIG. **1A** is a schematic diagram of an imaging system **100** according to embodiments of the present disclosure. The system **100** may include an intraluminal ultrasound imaging device **110**, a control and processing system **130** (for example, a console including a computer), and a patient interface module (PIM) **131** extending between the device **110** and the control and processing system **130**.

[0032] The ultrasound imaging device **110** may include a catheter **101**, which is shown in more detail in FIG. **1B**. The catheter **101** may include one or more flexible elongate members sized and shaped, structurally arranged, and/or otherwise configured to be positioned within a body lumen of a patient. In some embodiments, the catheter **101** includes an ultrasound imaging assembly **102**, a catheter body or shaft **201**, a catheter cable **203**, a handle **120**, a conduit **124**, a connector **209**, and one or more printed circuit board assemblies (PCBAs) **207**. The catheter cable **203** may have a small diameter configuration and a low profile that is sized to be passed or sneaked through a catheter shaft **201**, the handle **120**, and/or the conduit **124**. The cable **203** may be electrically and/or mechanically coupled to the ultrasound imaging assembly **102** at the distal portion of the catheter shaft **201**, as well as the PCBA **207** at the proximal portion of the catheter **101**.

[0033] In some embodiments, one or both of the catheter body/shaft **201** and catheter cable **203** may be referred to as a flexible elongate member. The catheter shaft **201** is sized and shaped, structurally arranged and/or otherwise configured to be positioned within a body lumen of a patient (e.g., vasculature such as blood vessels or chambers of the heart). Respective portions of the catheter cable **203** extend within the catheter shaft **201**, the handle **120**, the conduit **124**, and the connector **209**. The imaging assembly **102** may be attached at a distal end of the catheter shaft **201**. The catheter shaft **201** may include a lumen that the catheter cable **203** may pass through. The proximal end **204** shown in FIG. **2** of the catheter shaft **201** may be attached to the handle **120**, for example, by a resilient strain reliever. The handle **120** may be used for manipulation of the ultrasound imaging device **110** and manual control of the ultrasound imaging device **110**. The ultrasound imaging device **110** may include an imaging assembly **102** with ultrasound transducer elements and associated circuitry. The handle **120** may include actuators **116**, a clutch **114**, and other steering control components for steering the ultrasound imaging device **110**. The steering may include deflecting the distal end of the catheter cable **203**, as described in greater details herein.

[0034] The catheter cable **203** may pass through one or more of the catheter shaft **201**, handle **120**,

conduit **124**, and connector **209**. In some embodiments, during assembly, the catheter cable **203** is sneaked through a lumen within the catheter body **201**, handle **120**, and conduit **124**. In some embodiments, the conduit **124** is a component distinct from the cable **203**. For example, the conduit can be a tubing within which the cable **203** extends. In other embodiments, the conduit **124** can be a coating defining an exterior surface of the cable **203**. The coating can strengthen the cable **203** for exposure to direct contact and/or handling by an operator of the catheter **101**. The catheter cable **203** may be terminated at a PCBA **207** within the connector **209**. The catheter cable **203** may be electrically and mechanically coupled to the imaging assembly **102** and may include a plurality of electrical wires.

[0035] The handle **120** may be connected to the conduit **124** via another strain reliever. The conduit **124** may be configured to provide suitable configurations for interconnecting the control and processing system **130** and the monitor **132** to the imaging assembly **102**. The control and processing system **130** may be used for processing, storing, analyzing, and manipulating data, and the monitor **132** may be used for displaying obtained signals generated by the imaging assembly **102**. The control and processing system **130** can include a processor circuit with one or more processors in communication with memory. The memory can be a non-transitory computer readable storage medium. The memory can store instructions or program code that, when executed by the processor, causes the processing circuit to perform one or more functions described herein. The control and processing system **130** may be operable to facilitate the features of the intraluminal imaging system **100** described herein. For example, a processor can execute computer readable instructions stored on the non-transitory tangible computer readable medium. The control and processing system **130** can include one or more input devices, such as keyboards and any suitable command control interface device. The monitor **132** may be any suitable display device, such as liquid-crystal display (LCD) panel or the like.

[0036] In operation, a physician or a clinician may advance the catheter **101** into a lumen, such as a blood vessel, body lumen, or portion of a heart anatomy. By controlling actuators **116** and/or the clutch **114** on the handle **120**, the physician or clinician may steer the catheter **101** to a position near the area of interest to be imaged. For example, one actuator may deflect the imaging assembly **102** and a distal end of the catheter cable **203** in a left-right plane and the other actuator may deflect the imaging assembly **102** and the distal end of the catheter cable **203** in an anterior-posterior plane. The clutch **114** may provide a locking mechanism to lock the positions of the actuators **116** and in effect lock the deflection of the imaging assembly **102** while imaging the area of interest.

[0037] The imaging process may include activating the ultrasound transducer elements on the imaging assembly **102** to produce ultrasonic energy. A portion of the ultrasonic energy is reflected by the area of interest and the surrounding anatomy, and the ultrasound echo signals are received by the ultrasound transducer elements. The conduit **124** may be used to transfer the received echo signals to the control and processing system **130** where the ultrasound image is reconstructed and displayed on the monitor **132**. In some embodiments, the processing system **130** can control the activation of the ultrasound transducer elements and the reception of the echo signals. In some embodiments, the control and processing system **130** and the monitor **132** may be part of a same system.

[0038] While some embodiments of the present disclosure refer to an imaging device, an ultrasound imaging device, or an intraluminal imaging device, it is understood that the ultrasound imaging device **110** and the system **100** generally may be used to image vessels, structures, lumens, and/or any suitable anatomy/tissue within a body of a patient including any number of anatomical locations and tissue types, including without limitation, organs including the liver, heart, kidneys, gall bladder, pancreas, lungs; ducts; intestines; nervous system structures including the brain, dural sac, spinal cord and peripheral nerves; the urinary tract; as well as valves within the blood, chambers or other parts of the heart, and/or other systems of the body. In addition to natural structures, the imaging device **110** may be used to examine man-made structures such as, but

without limitation, heart valves, stents, shunts, filters and other devices. For example, the ultrasound imaging device **110** can be positioned within fluid filled or surrounded structures, both natural and man-made, such as within a body of a patient. The vessels, structures, lumens, and anatomy/tissue can include a blood vessel, such as an artery or a vein of a patient's vascular system, including cardiac vasculature, peripheral vasculature, neural vasculature, renal vasculature, and/or any suitable lumen inside the body.

[0039] The system **100** can be referenced as an imaging system, ultrasound imaging system, intraluminal imaging system, and/or combinations thereof. Although the present disclosure refers to ICE catheters, any suitable intraluminal imaging device is contemplated, such as an intravascular ultrasound (IVUS) device, an optical coherence tomography (OCT) device, an intracardiac echocardiography (ICE) device, a transesophageal echocardiography (TEE) device, an intravascular photoacoustic (IVPA) imaging device, and/or any suitable internal imaging device. Intraluminal devices with flexible elongate members such as catheters, guide wires, and/or guide catheter are contemplated.

[0040] The system **100** may be utilized in a variety of applications such as transseptal punctures, left atrial appendage closures, atrial fibrillation ablation, and valve repairs and can be used to image vessels and structures within a living body. Although the system **100** is described in the context of intraluminal imaging procedures, the system **100** is suitable for use with any catheterization procedure. In addition, the imaging assembly **102** may include any suitable physiological sensor or component for diagnostic, treatment, and/or therapy. For example, the imaging assembly can include an imaging component, an ablation component, a cutting component, a morcellation component, a pressure-sensing component, a flow-sensing component, a temperature-sensing component, and/or combinations thereof. In some embodiments, the intraluminal imaging system **100** is used for generating two-dimensional and three-dimensional images.

[0041] Referring back to FIG. 1A, the PIM **131** may provide a physical and electrical connection between the ultrasound imaging device **110** and the control and processing system **130**. Some embodiments of the present disclosure omit the PIM **131**. In other embodiments, the PIM **131** is communicatively interposed between the ultrasound imaging device **110** and the processing system **130**. In some instances, the PIM **131** can be referenced as a patient interface cable. For example, a proximal connector **209** of the ultrasound imaging device **110**, a distal connector of the PIM, and/or a proximal connector of the PIM may be configured to couple the ultrasound imaging device **110**, the PIM **131**, and the control and processing system together mechanically and electrically. The system **100** may include a connector junction **111** comprising a proximal connector **209** of the ultrasound imaging device **110** and the distal connector of the PIM **131**.

[0042] In some embodiments, the control and processing system **130** may include one or more computers, processors, and/or computer systems. The control and processing system **130** may also be referred to as a console. In some embodiments, the PIM **131** is in mechanical and electrical communication with the control and processing system **130**, such that the electrical signals are transmitted to the ultrasound imaging device **110** through the PIM **131** and to the control and processing system **130**. The control and processing system **130** may include one or more processors and/or memory modules forming a processing circuit or processor circuit that may process the electrical signals, and generate and output a graphical representation of the imaging data on the monitor **132** (e.g., an ultrasound image). One or more electrical conductors of the ultrasound imaging device **110** and PIM **131** may facilitate communication between the control and processing system **130** and the ultrasound imaging device **110**. For example, a user of the control and processing system **130** may control imaging using the ultrasound imaging device **110** via a control interface **134** of the control and processing system **130**. Electrical signals representative of commands from the control and processing system **130** may be transmitted to the ultrasound imaging device **110** via connectors and/or cables in the PIM **131** and the ultrasound imaging device **110**. The control and processing system **130** may be transportable and may include wheels or other

devices to facilitate easy transportation by a user.

[0043] In some embodiments, the one or more components of the ultrasound imaging device **110** may be disposable components. For example, a user, such as a physician, may obtain the catheter **101** and/or the ultrasound imaging device **110** in a sterilized packaging. In some embodiments, the ultrasound imaging device **110** may be disposed after a single use. In other embodiments, the ultrasound imaging device **110** can be sterilized and/or re-processed for more than one use. The PIM **131** may be a re-usable component that is used in multiple procedures. For example, the PIM **131** can be cleaned between individual procedures, such as being treated with disinfectants to kill bacteria. In some embodiments, the PIM **131** may not be required to be sterilized before a medical procedure. For example, the PIM **131** can be sufficiently spaced from the patient such that use of a non-sterile PIM **131** is safe for the patient. The sterile-nonsterile connection at the connector assembly **111** between the ultrasound imaging device **110** and the PIM **131** may allow for a safe operating environment while saving costs by allowing expensive equipment to be reused.

[0044] FIG. **2** is a perspective view of the catheter cable **203** described above with respect to FIG. **1B**. The catheter cable **203** is a flexible elongate body **206** including a plurality of communication cables allowing communication of imaging data and/or command signals between the processing system **130** and the catheter **101**. The communication cables can be electrical wires. An individual electrical wire can include a bare single conductor surrounded by one or more insulating layer(s) and/or shielding layer(s). An individual electrical wire can also be a coaxial wire including a center conductor surrounded by one or more insulating layer(s) and a conductive shield layer also surrounded by one or more insulating or protective layer(s). The plurality of electrical wires collectively can be surrounded by one or more insulating layers(s) and/or shielding layer(s). An insulating layer may be formed of any suitable materials, such as a plastic or a polymer in some instances. A shield layer may be formed of any suitable material, such as a metal in some instances. For example, a woven layer **211**, such as an RFI braid can surround the electrical wires. The cable **203** extends between the ultrasound imaging assembly **102** disposed at a distal portion **202** and PCBA **207** at a proximal portion **204**. The flexible elongate body **206** extends between the distal end **202** and the proximal end **204**. In some embodiments, the imaging assembly **102** is electrically and/or mechanically coupled (e.g., adhered or bonded) to distal end **202**. During manufacturing, the imaging assembly **102** may be coupled to the catheter cable **203**, prior to the cable **203** being sneaked through the catheter body or shaft. In some embodiments, the catheter cable **203** is about 4 feet long. In other embodiments, the catheter cable **203** is between 1 and 6 feet long or between 3 and 5 feet long, and/or other suitable values both larger and smaller.

[0045] FIG. **3A** shows a cross-sectional view of a catheter shaft **201**. The catheter shaft **201** is sized and shaped, structurally arranged, and/or otherwise configured to be positioned within the body lumen of a patient during the imaging procedure. The catheter cable **203** (as shown in FIGS. **2** and **3B**) may be configured to be disposed within an internal lumen **332** of the catheter shaft **201**. The catheter shaft **201** may include a number of pullwire lumens **336** disposed within the catheter shaft **201**. Pull wires positioned within the lumens **336** control movement (deflection of the distal tip) of the distal portion of the catheter shaft and/or the imaging assembly **102**. In some embodiments, the catheter shaft **201** has an outer diameter between approximately 1 mm and approximately 3 mm, including values both larger and smaller. In an exemplary embodiment, the catheter shaft **201** has an outer diameter of about 1.422 (+/-0.025 mm).

[0046] FIG. **3B** shows a cross-sectional view of a catheter cable **203**. The catheter cable **203** (e.g., the PCBA **207** and the flexible elongate body **206**) may be sneaked or passed through the catheter shaft **201** during assembly. The PCBA **207** can be configured to directly or indirectly interface with a user console. For example, the PCBA **207** can be in direct or indirect communication with the console or processing system **130** and/or the PIM **131** (FIG. **1A**). In some embodiments, the catheter cable **203** has a diameter between approximately 1 mm and approximately 3 mm, including values both larger and smaller. In an exemplary embodiment, the catheter cable **203** has a

diameter of about 1.3 mm ( $\pm 0.07$  mm). In some embodiments, the catheter cable **203** may include a polymer layer **342**, a shielding layer **344**, and a number of electrical wires **346**. The electrical wires **346** may be disposed within the shielding layer **342** which may be disposed within the polymer layer **342**. The electrical wires **346** may be used to communicate signals from the imaging assembly to the proximal end **204**, and ultimately to the processing system **130**. In some embodiments, the shield layer **342** can be the woven layer **211** disposed around the polymer layer **342**, as shown in FIG. 2. The electrical wires **346** connect the imaging assembly **102** and the proximal connector **209** (e.g., PCBA **207**).

[0047] FIG. 4 is a perspective view of the imaging assembly **102** according to embodiments of the present disclosure. The imaging assembly **102** is positioned at the distal portion of the catheter shaft **201** after assembly. The imaging assembly **102** is also positioned at the distal portion of the cable **203**. The imaging assembly **102** may include an ultrasound transducer array **262** that includes a number of transducer elements and a micro-beam-former IC **304** that can be coupled to the transducer array **262**. The electrical wires **346** of the cable **203** are mechanically and electrically coupled to the imaging assembly **102**. In some examples, the electrical cable **203** is further coupled through an interposer **310** to the micro-beam-former IC **304**. In some examples the interposer **310** is connected to the micro-beam-former IC **304** through wire bonding **320**. The wires **346** of the cable **203** are directly or indirectly in communication with the transducer array **262**, the IC **304**, and/or the interposer **310**. In some embodiments, the interposer **310** may be a circuit board or any other suitable component. In some embodiments, the wires **346** comprises three groups: a first set of coaxial cables **402**, a second set of coaxial cables **404**, and a third set of standard single conductor cables **406**.

[0048] In some embodiments, the transducer array **262** includes ultrasound imaging transducers that are directly flip-chip mounted to the micro-beam-former IC **304**. The transmitters and receivers of the ultrasound imaging transducers are on the micro-beam-former IC **304** and are directly attached to the transducers. In some examples, a mass termination of the acoustic elements is done at the micro-beam-former IC **304**.

[0049] In some examples, the transducer array **262** includes more than 800 imaging elements and the electrical cable **203** includes a total of 12 signal lines or less. In some examples, the electrical cable **203** includes a total of 30 lines or less that includes the signal lines, power lines, and control lines. In some examples, the transducer array **262** includes a one-dimensional or two-dimensional array from between 32 to 1000 imaging elements. For example, the array can include 32, 64, 128, 256, 512, 640, 768, 812, or any other suitable number of imaging elements. For example, a one-dimensional array may have 32 imaging elements. A two-dimensional array may have 32, 64, or more imaging elements. In some examples, the number of signal lines is between 10 and 20, for example, 12 signal lines, 16 signal lines, or any other suitable number of signal lines. A one-dimensional array can be configured to generate two-dimensional images. A two-dimensional array can be configured to generate two-dimensional and/or three-dimensional images.

[0050] In some examples, the electrical cable **203** of the imaging assembly **102** is directly coupled to the micro-beam-former IC **304** of the imaging assembly **102**. In some embodiments, the micro-beam-forming IC **304** lies directly underneath the transducer array **262** and is electrically connected to them. The elements of the transducer array **262** may be piezoelectric or micromachined ultrasonic transducer (MUT) elements. In some examples, piezoelectric elements are attached to the IC **304** by flip-chip mounting of an assembly of acoustic layers that include sawing into individual elements. MUT elements may be flip-chip mounted as a unit or grown directly on top of the micro-beam-former IC **304**. In some examples, the cable bundle may be terminated to an interposer **310** of suitable material such as a rigid or flexible printed circuit assembly. The interposer **310** may then be connected to the micro-beam-forming IC **304** via any suitable means such as wire bondings **320**.

[0051] In some embodiments, the micro-beam-forming IC **304** and the interposer **310** are coupled

to an elongate acoustic backing material member **410**. For example, adhesive bonding may be used. In other embodiments, micro-beam-former IC **304** and interposer **310** may be mechanically fastened, solvent bonded, UV bonded, ultrasonically welded, or coupled using any other suitable method. Acoustic backing **410** may extend from the distal tip of the micro-beam-former IC **304** to the proximal end of interposer **310**.

[0052] A stiffening member **412** is coupled to acoustic backing **410**. Stiffening member **412** may be of similar shape to acoustic backing **410** and also extend from the distal tip of the micro-beam-former IC **304** to the proximal end of the interposer **310**. Stiffening member **412** provides additional rigidity and structure to imaging assembly **102**. Stiffening member **412** may be composed of any suitable material. In an exemplary embodiment, stiffening member **412** is made of stainless steel. In other embodiments, stiffening member may be made of electroless nickel plating, titanium, carbon fiber, magnesium, high specific strength steel, other alloy steels, or other suitable materials.

[0053] Interposer **310** houses various electrical components **420**. Electrical components **420** may be positioned between the distal termination of cables **326** and the wire bondings **320** and disposed on a top surface of interposer **310**. In some embodiments, electrical components **420** may be used to generate, convey, amplify, attenuate, record, or smooth signals to and from the transducer array **262**. Electrical components **420** may additionally be used for sensing one or more features of the physiology within which the intraluminal imaging device is positioned, such as the temperature. It is fully contemplated that electrical components **420** may serve any other function while positioned on imaging assembly **102**. Electrical components may consist of both passive and active components, including but not limited to resistors, capacitors, inductors, transistors, operational amplifiers, thermistors, or any other suitable electrical component.

[0054] Interposer **310** may also include grounding pad **430**. Grounding pad **430** may be a thin, conductive layer of material positioned on a top surface of interposer **310**. Grounding pad **430** can be disposed at the proximal portion of interposer **310**. A first set of coaxial cables **402** and a second set of coaxial cables **404** may be mechanically and electrically coupled to grounding pad **430**. This connection may be achieved by soldering an exposed shield layer of each coaxial cable (e.g., exposed shield layer **710** of FIG. 7) in first set **402** and second set **404** such that the cables are mechanically and electrically bonded to grounding pad **430** as will be discussed in more detail herein. In some embodiments, coaxial cables may be preferred over single conductor cables because they transmit fast-switching, high frequency signals more effectively and reduce noise. In some embodiments, the coaxial cables can be configured to carry control signals from the computer **130** to the ultrasound imaging assembly **102** and/or image data from the ultrasound imaging assembly **102** to the computer **130**.

[0055] FIG. 5 is a flow chart diagram of a method **500** of assembling an intraluminal imaging device **101** according to an embodiment of the present disclosure. The method **500** can include connecting the ultrasound imaging assembly **102** to the catheter cables **346**. As illustrated, method **500** includes a number of enumerated steps, but embodiments of method **500** may include additional steps before, after, or in between the enumerated steps. In some embodiments, one or more of the enumerated steps may be omitted, performed in a different order, or performed concurrently. The steps of method **500** can be carried out by a manufacturer of the intraluminal imaging device **101**, a manufacturer of a subassembly including the ultrasound imaging assembly **102** or the catheter cables **346**, and/or a manufacturer of any other component discussed in the present disclosure. Method **500** will be described with reference to FIGS. 6A-11, which are diagrammatic views of various components of the ultrasound imaging assembly **102** and cables **346** during various steps of manufacturing. For example, FIGS. 6A-11 illustrate assembly steps for various components of the device **110**, such as the connection between the ultrasound imaging assembly **102** and cables **346**.

[0056] At step **505**, method **500** includes obtaining an interposer **310** with a grounding pad **430**

disposed on a surface thereof, such as a top or superior surface. A diagrammatic top view of an exemplary interposer **310** with grounding pad **430** positioned at a proximal portion **620** of interposer **310**. is illustrated in FIG. **6A**, according to an embodiment of the present disclosure. Various components of the ultrasound imaging assembly **102** can be coupled to the interposer **310**. In some embodiments, interposer **310** may be a circuit board facilitating electrical communication between one or more components of the device **101**, such as the cables **346**, transducer array **262**, and/or the IC **304**. The circuit board can include conductive traces formed on a surface and/or an interior thereof. The interposer board can be formed of any suitable semiconductor material, such as a silicon (Si) substrate or a germanium (Ge) substrate. In some embodiments, interposer **310** may include a compound semiconductor such as silicon carbide (SiC), silicon germanium (SiGe), silicon germanium carbide (SiGeC). As shown in FIG. **6A**, conductive pads **602**, **604**, and **606** may be disposed on a top surface of interposer **310**. Electrical components **420** may also be disposed on a top surface of interposer **310**. Electrical components **420** may be of any particular type as discussed previously. Conductive pads **602**, **604**, and **606** may be positioned along a length of the interposer **310** between grounding pad **430** and electrical components **420**. Conductive pads **602** may have a proximal end **610** and a distal end **611**. Conductive pads **604** may also have a proximal end **612** and a distal end **613**. Additionally, grounding pad **430** may have a proximal end **451** and a distal end **450**. Electrical components **420** may positioned at a location distally from conductive pads **602**, **604**, or **606**. Wire bondings **320** may also be positioned on a top surface of interposer **310** and disposed at the distal portion **625** of the interposer **310**. The arrangement of the conductive pads **602**, **604**, and **606**, electrical components **420**, and/or the wire bondings **320** in the illustrated embodiment of FIG. **6A** is merely illustrative. Any of these features may also be positioned at any location on interposer **310**.

[0057] The interposer **310** can include dimensions **352**, **354**, and **356** which may be of any suitable size. The dimension **352** can be a width, the dimension **354** can be a length, and the dimension **356** can be a depth. In some embodiments, the width **352** of interposer **310** may be between approximately 1 mm and approximately 3 mm, including values both larger and smaller. In an exemplary embodiment, the width **352** of interposer **310** is about 2.31 mm (+/-0.100 mm). In some embodiments, the length **354** of interposer **310** may be between approximately 8 mm and approximately 15 mm, including values both larger and smaller. In an exemplary embodiment, the length **354** of interposer **310** is about 11.5 mm (+/-0.100 mm). The depth **356** of interposer **310** may be between approximately 0.10 mm and approximately 0.40 mm, including values both larger and smaller. In an exemplary embodiment, the depth **356** of interposer **310** is about 0.25 mm (+/-0.100).

[0058] The grounding pad **430** can include dimensions **432**, **434**, and **436**, which may be of any suitable size. The dimension **434** can be a width, the dimension **432** can be a length, and the dimension **436** can be a depth. Generally, the width **434** of grounding pad **430** is similar to or less than the width **352** of interposer **310**. Width **434** may be the same as width **352**. Length **432** of grounding pad **430** may generally be the same as length **712** of the exposed shield layer **710** of coaxial cables **700** shown in FIG. **7** and as will be discussed in more detail herein. The length **432** of grounding pad **430** may be of any suitable length as to allow room for other components on interposer **310** and create a necessary bonding between cables **346** and interposer **310**. In some embodiments, the length **432** of grounding pad **430** may be between approximately 0.5 mm and approximately 1.0 mm, including values both larger and smaller. In an exemplary embodiment, the length **432** of grounding pad **430** is about 0.69 mm (+/-0.050 mm). In some embodiments, the width **434** of grounding pad **430** may be between approximately 1.0 mm and approximately 3.0 mm, including values both larger and smaller. In an exemplary embodiment, the width **434** of grounding pad **430** is about 1.85 mm (+/-0.050 mm).

[0059] Referring now to FIG. **6B**, shown therein is a diagrammatic cross-sectional view of interposer **310** and grounding pad **430** along section line **6B-6B** in FIG. **6A**. As depicted in FIG.

6B, grounding pad **430** is a generally rectangular, thin conductive material disposed on interposer **310**. In some embodiments, the depth **436** of grounding pad **430** may be between approximately 0.010 mm and approximately 0.100 mm, including values both larger and smaller. In an exemplary embodiment, the depth **436** of grounding pad **430** is about 0.025 mm (+/-0.005 mm). In FIG. 6B, grounding pad **430** is depicted as being recessed into a corresponding cavity on a top surface of interposer **310** so as to make the top surface of grounding pad **430** level, flush, or continuous with the top surface of interposer **310**. This configuration represents one illustrative embodiment. It will be understood that grounding pad **430** may be disposed on top of interposer **310** without a recessed cavity such that grounding pad **430** protrudes from the upper surface of interposer **310** by a distance generally equal to depth **436** of grounding pad **430**. Alternatively, grounding pad **430** may be recessed into interposer **310** by a depth greater than depth **436** such that the upper surface of grounding pad **430** lies beneath the upper surface of interposer **310**. The exact orientation of grounding pad **430** in relation to interposer **310** may be of any particular type.

[0060] Referring again to FIG. 5, at step **510**, method **500** includes obtaining a first set of coaxial cables. An illustrative example of one embodiment of a coaxial cable that may be included in this first set of coaxial cables is shown in FIG. 7. Coaxial cable **700** includes four basic layers. It is fully contemplated that additional layers may be added to those shown in FIG. 7. For example, layers that provide additional strength, rigidity, protection, increased conductivity, or any other function may be added to coaxial cable **700**. For example, an inner shielding layer and an outer shield layer can be provided. The inner shield layer and/or the outer shield layer can be electrically and mechanically coupled to other components within ultrasound imaging assembly **102**. For simplicity's sake, only four layers are shown and discussed in detail herein.

[0061] A center conductor cable **720** may be positioned along the axial center of cable **700**. The center conductor **720** may comprise multiple conductive elongate wires twisted or otherwise coupled to one another or may comprise a single conductive elongate wire. The center conductor **720** may carry a signal from the control and processing system **130** to the ultrasound image assembly **102** or from the ultrasound imaging assembly **102** to the control and processing system **130**. Coaxial cable **700** may be provided in such a way that layers of cable **700** are removed to expose a distal portion of center conductor **720** of a length **722**. Cable **700** may be obtained with this distal portion of center conductor **720** already exposed. The length **722** of the distal portion of center conductor **720** may be of any suitable length. Length **722** will generally be of similar dimensions to conductive pads **602** or **604** so as to make electrical and mechanical bonding of the distal end of center conductor **720** to conductive pad **602** or **604** possible as will be discussed in more detail herein.

[0062] An insulation layer **715** may be disposed around center conductor **720**. The insulation layer **715** may extend primarily from the proximal to distal ends of center conductor **720**. A purpose of insulation layer **715** may be to insulate center conductor **720** from other layers of cable **700** and from the surrounding environment such that center conductor **720** is not in electrical communication with any elements except those intended at either end of center conductor **720**. As previously mentioned, insulation layer **715** is not disposed around a distal portion of center conductor **720** to allow electrical and mechanical bonding of the distal end of center conductor **720** to a conductive pad **602** or **604**. As shown in FIG. 7, outer layers of cable **700** are further removed to expose a length **717** of insulation layer **715**. Length **717** may be of any suitable length. Generally, length **717** may be approximately equal to the distance from the proximal end **610** of conductive pads **602** to the distal end **450** of grounding pad **430** for first set of coaxial cables **700a**. For a second set of cables **700b**, length **717** is approximately equal to the distance from the proximal end **612** of conductive pads **604** to the distal end **450** of grounding pad **430**, further accounting for any displacement of coaxial cables **700b** as a result of being positioned over cables **700a** as will be discussed herein.

[0063] A shield layer **710** may be positioned around insulation layer **715** extending primarily from

the proximal to distal ends of insulation layer **715**. A purpose of shield layer **710** may be to establish an electrical ground relative to center conductor **720** in electrical communication with control and processing system **130** and ultrasound imaging assembly **102**. Like the exposed distal portion of center conductor **720**, and the exposed distal portion of insulation layer **715**, an outer layer or jacket **705** of coaxial cable **700** is partially removed to expose a distal portion of shield layer **710**. As mentioned previously, the length **712** of this exposed distal portion of shield layer **710** is generally of the same or similar length as the width **432** of grounding pad **430**. However, the length **712** of the exposed distal portion of shield layer **710** may be of any suitable length.

[0064] An outer layer or jacket **705** is disposed around the shield layer **710** extending primarily from the proximal to distal ends of shield layer **710**. A purpose of jacket **705** may be to insulate shield layer **710** from the general environment and ensure that shield layer **710** is only in electrical communication with the control and processing system **130** and the ultrasound imaging assembly **102**. As previously stated, jacket **705** is removed to expose a distal portion of shield layer **710** of a length **712** as shown in FIG. 7.

[0065] Referring again to FIG. 5, at step **515**, method **500** includes mechanically and electrically coupling center conductors **720a** of a first set of coaxial cables **700a** to conductive pads **602** on interposer **310**. As shown in FIG. 8A, a first set of coaxial cables **700a** are substantially similar to one another. FIG. 8A depicts eight such coaxial cables positioned on interposer **310**, however, it is fully contemplated that any number of coaxial cables **700a** may be included in this first set. A first set of coaxial cables **700a** may include only one coaxial cable **700a**, two, four, eight, sixteen, or more of coaxial cables **700a**.

[0066] As shown in FIG. 8A, the center conductor **720a** of each coaxial cable **700a** is mechanically and electrically bonded to a corresponding conductive pad **602** positioned on interposer **310**. Because eight coaxial cables **700a** are shown in FIG. 8A, interposer **310** has eight conductive pads **602** positioned on interposer **310** near grounding pad **430**. As mentioned, this number is merely illustrative. An electrical and mechanical bond may be achieved by any suitable method, including but not limited to using solder.

[0067] Referring again to FIG. 5, at step **520**, method **500** includes soldering shield layers **710a** of coaxial cables **700a** to grounding pad **430**. In some embodiments a mass of solder **810** as shown in FIG. 8B is applied such that shield layers **710a** of coaxial cables **700a** are mechanically and electrically bonded to grounding pad **430**. As each shield layer **710a** of coaxial cables **700a** is to carry an equivalent electrical ground respective to the signal carried by each center conductor **720a** of coaxial cables **700a**, each shield layer **710a** of coaxial cables **700a** may be in electrical communication with each other and with grounding pad **430**. Grounding pad **430** provides an electrical ground for the plurality of coaxial cables **700a** and the plurality of coaxial cables **700b** as described in more detail herein. In addition to establishing an electrical connection, between shield layers **710a**, solder mass **810** provides a mechanical connection between coaxial cables **700a** and interposer **310** such that the connection between catheter cables **346** and ultrasound image assembly **102** is significantly stronger than traditional methods.

[0068] FIG. 8B is a cross-sectional view of interposer **310** along section line 8B-8B in FIG. 8A, with first row of coaxial cables **700a** connected according to an embodiment of the present disclosure. Eight coaxial cables **700a** are shown disposed over the upper surface of grounding pad **430** as previously mentioned. At the axial center of each coaxial cable is shown center conductor **720a**. Positioned around each center conductor **720a** is insulation layer **715a**. Positioned around each insulation layer **715a** is shown shield layer **710a**. Between insulation layer **715a** and shield layer **710a** is shown a layer of solder **812** that passes through shield layer **710a** during the soldering process. Solder **812** may be a part of or substantially similar to solder mass **810**. Shield layer **710a** may be constructed out of a braided conductive material so as to allow solder to pass through the layer. Alternatively, shield layer **710a** may be constructed of any suitable conductive material that allows solder to pass through the layer. In still other embodiments, shield layer **710a** may not be

porous at all and not allow solder to pass through the layer but still provides a strong mechanical connection. Shield layer **710a** is at least semi-porous to allow solder to pass through the layer as the resulting intertwined connection ensures a stronger mechanical connection between solder mass **810** and shield layer **710a** and subsequently between first set of coaxial cables **700a** and interposer **310**.

[0069] Without the solder mass as described in the present application, the connection between the distal tip of the ICE catheter, with ultrasound imaging assembly, and more proximal portions of the ICE catheter, such as the catheter shaft, are relatively weaker. For example, the distal tip and the catheter shaft are attached at a polymer-polymer interface between a polymer housing of the distal tip and the polymer forming the catheter shaft. The center conductors of the cables are also attached to their respective conductive pads for electrical communication. However, these connections are weaker and cannot handle larger tensile forces (e.g., at least 15 N).

Advantageously, mechanically coupling the shield layers of the plurality of coaxial cables to the grounding pad **430** via the solder mass as described in the present disclosure provides a strong mechanical connection between electrical wires **346** and ultrasound imaging assembly **102**. The solder mass connection thus provides a stronger point of connection between the distal tip of the ICE catheter and more proximal portions (e.g., stronger than the polymer-polymer interface and the center conductor-conductive pad connection). The presently disclosed method and apparatus advantageously satisfy minimum peak tensile force requirements in ISO 10555, among other national and international requirements of peak tensile force for intra-cardiac echocardiography (ICE) catheters that have a catheter shaft diameter larger than 1.85 mm. In some embodiments, the catheter shaft **201** (FIG. 3A) has a diameter of 9 Fr (3 mm). For example, the solder mass **810** is configured to support a tensile load of at least 15 N. For example, the solder mass **810** is configured to support a tensile load between approximately 1 N and approximately 60 N, between approximately 15 N and approximately 60 N, and/or other values both larger and smaller.

[0070] Solder mass **810** takes the same general shape of grounding pad **430** as the solder will wick to all conductive surfaces it comes in contact with and avoid mechanical bonding with non-conductive surfaces. For example, the solder mass can have a generally shaped as a rectangular or polygonal prism, with a generally rectangular or polygonal cross-sectional profile. In some embodiments, interposer **310** is constructed of a non-conductive material such that solder mass **810** only bonds with grounding pad **430** and shield layers **710a**. In addition, insulation layer **715a** and jacket **705a** of coaxial cables **700a** are constructed of non-conductive material such that solder mass **810** does not electrically or mechanically bond with these elements as well resulting in solder mass **810** retaining a shape generally similar to grounding pad **430**.

[0071] Referring again to FIG. 5, at step **525**, method **500** includes obtaining a second set of coaxial cables **700b**. The second set of coaxial cables **700b** may be substantially similar to first set of coaxial cables **700a**. An illustrative example of one embodiment of a coaxial cable **700** in this second set **700b** is shown in FIG. 7.

[0072] Second set of coaxial cables **700b** may differ from the first set of coaxial cables **700a** in regards to length **717** of insulation layer **715**. Referring again to FIG. 6A, length **717** of insulation layer **715** of coaxial cables **700b** may be the same general length as the distance between the proximal end **604b** of conductive pads **604** and the distal end **430a** of grounding pad **430**. The length **717** of insulation layer **715** of coaxial cables **700b** may be slightly longer than this distance due to an offset of the location of cables **700b** as cables **700b** may be positioned above first set of coaxial cables **700a**.

[0073] Referring again to FIG. 5, at step **530**, method **500** includes mechanically and electrically coupling center conductors **720b** of a second set of coaxial cables **700b** to conductive pads **604** on interposer **310**. As shown in FIG. 9A, a second set of coaxial cables **700b** are positioned over the top of a first set of coaxial cables **700a**. Like cables **700a**, coaxial cables **700b** are each substantially similar to one another. FIG. 9A depicts eight such coaxial cables **700b** positioned on

interposer **310** and eight coaxial cables **700a** positioned beneath coaxial cables **700b**. FIG. **9A** therefore shows 16 coaxial cables **700** positioned on interposer **310**, however, it is fully contemplated that any number of coaxial cables **700** may be included. Similarly, a second set of coaxial cables **700b** may include only one coaxial cable **700b**, two, four, eight, sixteen, or more of coaxial cables **700b**.

[0074] As shown in FIG. **9A**, center conductor **720b** of each coaxial cable **700b** is mechanically and electrically bonded to a corresponding conductive pad **604** positioned on interposer **310**. Because eight coaxial cables **700b** are shown in FIG. **9A**, interposer **310** shows eight conductive pads **604**. As mentioned, this number is merely illustrative. In some embodiments, conductive pads **602** may be positioned between conductive pads **604** and grounding pad **430**. This orientation allows the second set of coaxial cables **700b** to be positioned over the first set of coaxial cables **700a** so as to create two rows of coaxial cables as shown in FIG. **9B**. Arranging coaxial cables **700a** and **700b** over one another in two rows with conductive pads **602** and **604** arranged in two corresponding rows allows all cables **346** to be terminated and mechanically and electrically coupled to interposer **310** in less space on interposer **310** than traditional methods. This configuration of two or more rows of coaxial cables as disclosed in the present invention makes the entire ultrasound imaging assembly **102** shorter which allows for easier maneuvering of the intraluminal catheter assembly **101** within the heart, vasculature, or other lumen within a patient. A mechanical and electrical bond between center conductors **720b** and conductive pads **604** may be achieved by any suitable method, including but not limited to using solder.

[0075] Referring again to FIG. **9A**, the proximal portions of first set of coaxial cables **700a** and second set of coaxial cables **700b** are depicted as being offset from one another, such that second set of coaxial cables **700b** appear to terminate at some location distal of the proximal termination of first set of coaxial cables **700a**. Although this orientation may be utilized in an embodiment of the present disclosure, this staggering of the proximal termination of coaxial cables is depicted for primarily pedagogical purposes to show that first set of coaxial cables **700a** are positioned beneath second set of coaxial cables **700b** despite second set of coaxial cables **700b** completely obscuring the view of first set of coaxial cables **700a** in FIG. **9A**.

[0076] Referring again to FIG. **5**, at step **535**, method **500** includes soldering shield layers **710b** of coaxial cables **700b** to grounding pad **430**. In some embodiments, the mass of solder **810** which was applied to shield layers **710a** of first set of coaxial cables **700a** is expanded to further encompass shield layers **710b** of second set of coaxial cables **700b** and shield layers **710a** of first set of coaxial cables **700a**. In this manner, shield layers **710b** of second set of coaxial cables **700b** are mechanically and electrically bonded to both shield layers **710a** of first set of coaxial cables **700a** and grounding pad **430**. As each shield layer **710a** and **710b** of coaxial cables **700a** and **700b** carry an equivalent electrical ground respective to the signal carried by each center conductor **720a** and **720b** of coaxial cables **700a** and **700b**, each shield layer **710a** and **710b** of coaxial cables **700a** and **700b** may be in electrical communication with each other and with grounding pad **430**. In addition to establishing an electrical connection between shield layers **710a**, **710b**, and grounding pad **430**, solder mass **810** provides a mechanical connection between coaxial cables **700a**, **700b**, and interposer **310** such that the connection between catheter cables **346** and ultrasound image assembly **102** is significantly stronger than traditional methods.

[0077] FIG. **9B** is a cross-sectional view of interposer **310** along section line **9B-9B** in FIG. **9A**, with a first row of coaxial cables **700a** and a second row of coaxial cables **700b** bonded to grounding pad **430** according to an embodiment of the present disclosure. Eight coaxial cables **700a** are shown disposed along the upper surface of grounding pad **430** as previously mentioned. Further, eight coaxial cables **700b** are shown disposed along the upper surfaces of the first set of coaxial cables **700a**. Similar to first set of coaxial cables **700a**, second set of coaxial cables **700b** may include a shield layer **710b** that is at least semi-porous so as to allow a small amount of solder **812** to pass through shield layer **710b** and ensure a stronger mechanical connection between solder

mass **810** and shield layer **710b** and subsequently between second set of coaxial cables **700b**, first set of coaxial cables **700a**, and interposer **310**.

[0078] Solder mass **810** still retains the same general shape of grounding pad **430** as the solder will wick to all conductive surfaces it comes in contact with and avoid mechanical bonding with non-conductive surfaces. In some embodiments, interposer **310** is constructed of a material such that solder mass **810** only bonds with grounding pad **430** and shield layers **710a** and **710b**. In addition, insulation layer **715b** and jacket **705b** of coaxial cables **700b** are constructed of non-conductive material such that solder mass **810** does not electrically or mechanically bond with these elements as well. Solder mass **810** has a width similar to width **434** of grounding pad **430** and a length similar to length **432** of grounding pad **430**. The vertical depth of solder mass **810** is largely dependent on the dimensions and positions of coaxial cables **700a** and **700b**. The vertical depth of solder mass **810** may be between approximately 0.25 mm and approximately 0.75 mm, including values both larger and smaller. In an exemplary embodiment, the vertical depth of solder mass **810** is about 0.50 mm (+/-0.10 mm). The solder mass completely or partially surrounds a perimeter of each of the coaxial cables. For example, in a cross-section of the coaxial cables, the solder mass can completely or partially surround one or more of the coaxial cables. The solder mass **810** extends within lateral and vertical space between adjacent coaxial cables. For example, solder mass **810** will fill space between coaxial cables positioned left and right of each other within the same set and will fill space between coaxial cables positioned above and below each other within first set of coaxial cables **700a** and second set of coaxial cables **700b**. In the exemplary embodiment disclosed, solder mass **810** is positioned between the first row of coaxial cables **700a** and the second row of coaxial cables **700b**. Solder mass **810** is also positioned between each coaxial cable within first set of coaxial cables **700a** and between each coaxial cable within second set of coaxial cables **700b**. Solder mass **810** is also positioned between first set of coaxial cables **700a** and grounding pad **430**.

[0079] Referring again to FIG. 5, method **500** describes mechanically and electrically bonding center conductors **720a** to conductive pads **602** before mechanically and electrically bonding shield layers **710a** to grounding pad **430** in steps **515** and **520**. Similarly, method **500** describes mechanically and electrically bonding center conductors **720b** to conductive pads **604** before mechanically and electrically bonding shield layers **710b** to grounding pad **430** in steps **530** and **535**. However, this particular order of connecting center conductors **720a** or **720b** of coaxial cables **700a** or **700b** before connecting shield layers **710a** or **710b** to grounding pad **430** is not required. Shield layers **710a** or **710b** may be mechanically and electrically bonded to grounding pad **430** before center conductors **720a** or **720b** are bonded to conductive pads **602** or **604**.

[0080] Referring again to FIG. 5, at step **540**, method **500** includes obtaining a set of single conductor cables **1000**. Single conductor cables **1000** may be substantially similar to single conductor cables well known to those skilled in the art. FIG. **10A** shows a third set of five such single conductor cables positioned over first set of coaxial cables **700a** and second set of coaxial cables **700b**. In some embodiments, single conductor cables **1000** are primarily composed of two layers. It is contemplated, however, that additional layers may be added in other embodiments of the present invention which provide additional functionality. For example, layers that provide additional strength, rigidity, protection, increased conductivity, or any other function may be added to single conductor cables **1000**. As shown in FIG. **10A**, a single conductor **1010** extends along the inside of single conductor cables **1000**. Single conductor **1010** may be used to transmit signals from the control and processing system **130** to the ultrasound imaging assembly **102** or from the ultrasound imaging assembly **102** to the control and processing system **130**. For example, the single conductor **1010** can be a power line or thermistor sensing lead. In such embodiments, the single conductor **1010** can carry power signals or thermistor signals. The single conductor **1010** may be used for any number of other suitable applications as well.

[0081] Disposed around single conductor **1010** is an insulation layer **1005**. Insulation layer **1005** extends primarily from the proximal to distal ends of single conductor **1010**. A purpose of

insulation layer **1005** may be to insulate single conductor **1010** from the general environment and ensure that single conductor **1010** is only in electrical communication with the control and processing system **130** and the ultrasound imaging assembly **102** as may be intended. Insulation layer **1005** is removed to expose a distal portion of single conductor **1010** as shown in FIG. **10A**. Single conductor cables **1000** may be obtained in such a way as to expose the distal portion of single conductor **1010** or single conductor cables **1000** may alternatively be modified after obtaining cables **1000** such that single conductor **1010** is exposed as shown. Although the present disclosure depicts only two layers comprising single conductor cables **1000**, many additional layers serving different functions may be included in single conductor cables **1000** as previously mentioned.

[0082] Referring again to FIG. **5**, at step **545**, method **500** includes mechanically and electrically coupling single conductor cables **1000** to conductive pads **606** such that single conductor cables **1000** are positioned above first set of coaxial cables **700a** and second set of coaxial cables **700b**. As shown in FIG. **10A**, single conductor cables **1000** are substantially similar to one another. FIG. **10A** depicts five such single conductor cables **1000** positioned on interposer **310**, however, it is fully contemplated that any number of single conductor cables **1000** may be included in this set. Single conductor cables **1000** may include only one single conductor cable **1000**, two, four, eight, 16, hundreds or more of single conductor cables **1000**.

[0083] As shown in FIG. **10A**, single conductor **1010** of each single conductor cable **1000** is mechanically and electrically bonded to a corresponding conductive pad **606** positioned on interposer **310**. Because five single conductor cables **1000** are shown in FIG. **10A**, interposer **310** shows five corresponding conductive pads **606** positioned near conductive pads **604**. As mentioned, this number is merely illustrative. In some embodiments, conductive pads **602** are positioned between conductive pads **604** and grounding pad **430**, and conductive pads **604** are positioned between conductive pads **606** and conductive pads **602**. This orientation allows the second set of coaxial cables **700b** to be positioned over the first set of coaxial cables **700a**, and the single conductor cables **1000** to be positioned over both first set of coaxial cables **700a** and second set of coaxial cables **700b** as to create two rows of coaxial cables **700a** and **700b** and one row of single conductor cables **1000** as shown in FIG. **10B**. However, conductor pads **602**, **604**, and **606** may be positioned at any place on interposer **310**. A mechanical and electrical bond between single conductors **1010** and grounding pads **606** may be achieved by any suitable method, including but not limited to the use of solder, adhesive, insulating tape, wire gel connectors, grease filled connectors, rubber welding, heat shrinking, or any other suitable form of mechanical and electrical connection. It is also contemplated that single conductor cables **1000** need not be positioned above first set of coaxial cables **700a** and second set of coaxial cables **700b**. They may be positioned to either side of or beneath the coaxial cables in any suitable manner. It is noted that solder mass **810** which encapsulates first set of coaxial cables **700a** and second set of coaxial cables **700b** does not encapsulate single conductor cables **1000**. Because insulation layer **1005** extends all along single conductor **1010** except near the distal end of single conductor **1010** where it is bonded to conductive pad **606**, the solder of solder mass **810** does not wick to single conductor cables **1000**.

[0084] Referring again to FIG. **10A**, the proximal portions of first set of coaxial cables **700a**, second set of coaxial cables **700b**, and single conductor cables **1000** are depicted as being offset from one another, such that second set of coaxial cables **700b** appear to terminate at some location distal to the proximal termination of first set of coaxial cables **700a** and single conductor cables **1000** appear to terminate at some location distal to the proximal termination of second set of coaxial cables **700b**. Although this orientation may be utilized in an embodiment of the present disclosure, this staggering of the proximal termination of coaxial cables is depicted for primarily pedagogical purposes similar to the staggering of first set of coaxial cables **700a** and second set of coaxial cables **700b** as described in relation to FIG. **9A**. FIG. **10A** illustrates the presently disclosed apparatus after all previously mentioned coaxial and single conductor cables have been connected

to interposer **310**.

[0085] FIG. **10B** is a cross-sectional view of an interposer **310** along section line **10B-10B** in FIG. **10A**, with first row of coaxial cables **700a** and second row of coaxial cables **700b** connected and a third top row of center conductor cables **1000** positioned above first row of coaxial cables **700a** and second row of coaxial cables **700b** according to an embodiment of the present disclosure. Solder mass **810** envelopes first row of coaxial cables **700a** and second row of coaxial cables **700b** and mechanically and electrically couples them to grounding pad **430**. As previously mentioned, each coaxial cable within first set of coaxial cables **700a** comprises center conductor **720a**, insulation layer **715a**, and shield layer **710a**. Similarly, each coaxial cable within second set of coaxial cables **700b** comprises center conductor **720b**, insulation layer **715b**, and shield layer **710b**. Solder **812** which may be a portion of or substantially similar to solder mass **810** may be positioned between insulation layer **715a** and shield layer **710a** for first set of coaxial cables **700a** and between insulation layer **715b** and shield layer **710b** for second set of coaxial cables **700b**. As previously mentioned, single conductor cables **1000** each comprise a single conductor **1005** and an insulation layer **1010**. It is also noted that cables **346** as shown in FIG. **3B** are arranged in a circular manner within the catheter cable **203**. At a point proximal to the proximal portion **620** of interposer **310**, cables **346** being to transition their arrangement into the three or more rows as shown in FIG. **10B**. FIG. **10B** illustrates the presently disclosed apparatus after all previously mentioned coaxial and single conductor cables have been connected to interposer **310**. In some embodiments, the length and/or width of the solder mass **810** can be equal to the length **432** and/or width **434**, respectively, of the grounding pad **430**. In some embodiments, the length and/or width of the solder mass **810** can be less than the length **432** and/or width **434**, respectively, of the grounding pad **430**. The height of the solder mass **810** can depend on the volume of solder that is used to couple the ground padding **430**, the first set of coaxial cables **700a**, and the second set of coaxial cables **700b**.

[0086] FIG. **11** is a perspective view of an interposer **310** with catheter cables **346** comprising first set of coaxial cables **700a**, second set of coaxial cables **700b**, and third set of single conductor cables **1000**, mechanically and electrically coupled to corresponding conductive pads **602**, **604**, and **606** and grounding pad **430**. Solder mass **810** mechanically and electrically couples first set of coaxial cables **700a** and second set of coaxial cables **700b** to grounding pad **430**. Electrical components **420** may be positioned distally to grounding pads **602**, **604**, and **606**. FIG. **11** illustrates the presently disclosed apparatus after all previously mentioned coaxial and single conductor cables have been connected to interposer **310**.

[0087] It is noted that although the figures and associated description of the present disclosure depict three sets of cables **346**, a first set of coaxial cables **700a**, a second cables of coaxial cables **700b**, and a set of single conductor cables **1000** arranged in three rows, any number of sets of cables may be used. For example, only one set arranged in one row may be used and mechanically and electrically bonded to interposer **310** according to the embodiments of the present disclosure. Additionally, two, four, five, ten, or more sets of cables may be mechanically and electrically bonded to interposer **310** according to the present disclosure. Additionally, single conductor cables **1000** may be positioned in the same row or set as first set of coaxial cables **700a** or second set of coaxial cables **700b**. Single conductor cables may be in the same row or set as any other type of cable used in the present disclosure.

[0088] FIG. **12** is a schematic diagram of a processor circuit **150**, according to embodiments of the present disclosure. The processor circuit **150** may be implemented in the intraluminal ultrasound imaging device **110**, the control and processing system **130**, and/or the PIM **131** of FIG. **1A**. As shown, the processor circuit **150** may include a processor **160**, a memory **164**, and a communication module **168**. These elements may be in direct or indirect communication with each other, for example via one or more buses.

[0089] The processor **160** may include a central processing unit (CPU), a digital signal processor (DSP), an ASIC, a controller, an FPGA, another hardware device, a firmware device, or any

combination thereof configured to perform the operations described herein. The processor **160** may also be implemented as a combination of computing devices, e.g., a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration.

[0090] The memory **164** may include a cache memory (e.g., a cache memory of the processor **160**), random access memory (RAM), magnetoresistive RAM (MRAM), read-only memory (ROM), programmable read-only memory (PROM), erasable programmable read only memory (EPROM), electrically erasable programmable read only memory (EEPROM), flash memory, solid state memory device, hard disk drives, other forms of volatile and non-volatile memory, or a combination of different types of memory. In an embodiment, the memory **164** includes a non-transitory computer-readable medium. The memory **164** may store instructions **166**. The instructions **166** may include instructions that, when executed by the processor **160**, cause the processor **160** to perform the operations described herein with reference to the intraluminal ultrasound imaging device **110**, the control and processing system **130**, and/or the PIM **131** (FIG. **1A**). Instructions **166** may also be referred to as code. The terms “instructions” and “code” should be interpreted broadly to include any type of computer-readable statement(s). For example, the terms “instructions” and “code” may refer to one or more programs, routines, sub-routines, functions, procedures, etc. “Instructions” and “code” may include a single computer-readable statement or many computer-readable statements.

[0091] The communication module **168** can include any electronic circuitry and/or logic circuitry to facilitate direct or indirect communication of data between the intraluminal ultrasound imaging device **110**, the control and processing system **130** (including the monitor **132** and the control interface **134**), and/or the PIM **131**. In that regard, the communication module **168** can be an input/output (I/O) device. In some instances, the communication module **168** facilitates direct or indirect communication between various elements of the processor circuit **150** and/or the system **100**.

[0092] Persons skilled in the art will recognize that the apparatus, systems, and methods described above can be modified in various ways. Accordingly, persons of ordinary skill in the art will appreciate that the embodiments encompassed by the present disclosure are not limited to the particular exemplary embodiments described above. In that regard, although illustrative embodiments have been shown and described, a wide range of modification, change, and substitution is contemplated in the foregoing disclosure. It is understood that such variations may be made to the foregoing without departing from the scope of the present disclosure. Accordingly, it is appropriate that the appended claims be construed broadly and in a manner consistent with the present disclosure.

## Claims

1. An apparatus, comprising: an intracardiac echocardiography (ICE) catheter comprising: a catheter shaft comprising: a distal portion configured to be positioned within a heart of a patient; a first set of electrical wires; and a second set of electrical wires; and an ultrasound imaging assembly positioned at the distal portion of the catheter shaft, wherein the ultrasound imaging assembly is coupled to the first set of electrical wires and the second set of electrical wires, wherein the ultrasound imaging assembly comprises: a transducer array configured to obtain ultrasound data; and a solder mass surrounding the first set of electrical wires, wherein the solder mass comprises a first surface coupling the first set of electrical wires to the ultrasound imaging assembly and a different, second surface, wherein the second set of electrical wires is not surrounded by the solder mass, wherein the second set of electrical wires extends proximate to the second surface of the solder mass.
2. The apparatus of claim 1, wherein the first surface comprises a bottom surface of the solder

mass, wherein the second surface comprises an opposite, top surface of the solder mass.

3. The apparatus of claim 2, wherein the second set of electrical wires extends over top of the first set of electrical wires.
  4. The apparatus of claim 3, wherein the first set of electrical wires is arranged as at least one first row and the second set of electrical wires is arranged as second row.
  5. The apparatus of claim 1, wherein the ultrasound imaging assembly comprises a circuit board, wherein the circuit board comprises a grounding pad, wherein the first surface of the solder mass is coupled to the grounding pad.
  6. The apparatus of claim 5, wherein the circuit board comprises a first set of conductive pads distinct from the grounding pad, wherein the first set of electrical wires is coupled to the first set of conductive pads and to the grounding pad.
  7. The apparatus of claim 6, wherein the grounding pad is proximal of the first set of conductive pads along a length of the circuit board.
  8. The apparatus of claim 6, wherein the circuit board comprises a second set of conductive pads distinct from the grounding pad, wherein the second set of electrical wires is coupled to the second set of conductive pads.
  9. The apparatus of claim 8, wherein the first set of conductive pads is proximal of the second set of conductive pads along a length of the circuit board.
  10. The apparatus of claim 9, wherein the first set of electrical wires comprises a subset of electrical wires and a different subset of electrical wires, wherein the different subset of electrical wires extends over top of the subset of electrical wires.
  11. The apparatus of claim 10, wherein the first set of conductive pads comprises a subset of conductive pads and a different subset of conductive pads, wherein the subset of electrical wires is coupled to the subset of conductive pads and the different subset of electrical wires is coupled to the different subset of conductive pads.
  12. The apparatus of claim 11, wherein the subset of conductive pads is proximal of the different subset of conductive pads along the length of the circuit board.
  13. The apparatus of claim 1, wherein a portion of the solder mass is positioned between the first set of electrical wires and the second set of electrical wires.
  14. The apparatus of claim 1, wherein the first set of electrical wires comprises a first electrical wire type and the second set of electrical wires comprises a different, second electrical wire type.
  15. The apparatus of claim 14, wherein the first electrical wire type comprises a coaxial cable, and wherein the second electrical wire type comprises a single conductor cable.
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